PEG10 (Q86TG7): A Domesticated Retroviral Gag/Protease That Packages RNA, Signals Between Cells, and Builds the Placenta

Summary

PEG10 (Paternally Expressed Gene 10; UniProt **Q86TG7**; human chromosome **7q21**) is a protein-coding gene that originated by **domestication of a Ty3/gypsy LTR retrotransposon**. Its primary molecular function is that of a **retroviral Gag-like capsid protein**: it self-assembles into capsid-like **virus-like particles (VLPs)** that selectively bind and package RNA — preferentially its own mRNA and U/G-rich transcripts — and can bud from membranes into secreted particles that transfer mRNA cargo between cells. Through a **programmed −1 ribosomal frameshift** at a conserved slippery site/pseudoknot, PEG10 produces both a short **GAG (RF1)** product and a longer **GAG-POL (RF1/2)** fusion that carries a functional **viral aspartic protease** (DSG/DTG motif). This architecture — Gag capsid + Pol-like protease, generated by frameshifting — directly mirrors that of retroviruses and retrotransposons and has been confirmed at the structural level by X-ray crystallography.

Beyond its structural/RNA-packaging role, PEG10 acts as a **signaling adaptor and anti-apoptotic factor**. The GAG-like product PEG10-RF1 directly **binds and inhibits TGF-β-superfamily type I/II receptors** (ALK1/ACVRL1 and ALK5), and PEG10 physically associates with and antagonizes the **pro-apoptotic ubiquitin ligase SIAH1**, thereby suppressing cell death and promoting proliferation. It functions mainly in the **cytoplasm** (budding at membranes) and **extracellularly** (via secreted VLPs).

Physiologically, PEG10 is **essential and non-redundant for placentation**: knockout mice die at ~E10.5 from placental failure, and its viral protease activity is specifically required to maintain fetal placental capillary structure. Its oncofetal reactivation drives multiple cancers — most notably **hepatocellular carcinoma (HCC)**, where it stabilizes **GLUT1** and **HK2** mRNAs by blocking the Staufen1-mediated decay pathway to fuel aerobic glycolysis — and its escape from **UBQLN2** proteostatic control in neurons perturbs mRNA splicing in a **TDP-43-like**

manner, implicating it in ALS and related neurodegeneration. This report synthesizes 11 confirmed findings from 35 reviewed papers into a coherent mechanistic model of PEG10 as a domesticated Gag/protease RNA-packaging and intercellular-signaling protein.

Gene identity confirmation: The symbol PEG10 (HGNC:14005; synonyms EDR, MART2/MAR2, MEF3L1, RGAG3, KIAA1051e, SIRH1), the organism *Homo sapiens*, and the InterPro capsid/protease/reverse-transcriptase domains (IPR063300 PEG10-RTL1_C-capsid; IPR021109 Peptidase_aspartic_dom_sf; IPR063610 PRO_PEG10-RTL1) all align precisely with the literature synthesized here. There is no gene-symbol ambiguity.

Key Findings

Finding 1 — PEG10 is a Ty3/gypsy retrotransposon-derived, paternally imprinted gene essential for placentation

PEG10 is a paternally expressed (monoallelic) imprinted gene located within the **PEG10/SGCE imprinted cluster** at human chromosome 7q21, regulated by a germline differentially methylated region (gDMR). It is **conserved across therian mammals** (both eutherians and marsupials) and derives from a **Ty3/gypsy LTR retrotransposon**, retaining GAG- and POL-like domains. Its essential developmental role is established by direct genetic evidence: *Peg10* knockout mice show **early embryonic lethality at ~E10.5 owing to severe placental defects** (PMID: 16341224), a result independently corroborated in later work (PMID: 34559199). The same foundational study characterized PEG10 as one of the "actively transcribed Ty3/gypsy retrotransposon-derived genes with highly conserved DNA sequences and insertion sites," firmly establishing both its evolutionary origin and the deep conservation that signals functional domestication.

This finding anchors the entire investigation: PEG10 is not a parasitic mobile element but a **co-opted single-copy cellular gene** whose retroviral ancestry is the key to its biochemistry.

Finding 2 — PEG10 is a Gag-like RNA-binding protein that self-assembles into virus-like particles and packages its own mRNA

PEG10's N-terminal **GAG/capsid domain** promotes vesicle budding in a manner analogous to the HIV p24 Gag protein: "the Gag domain of PEG10 can promote vesicle budding similar to the HIV p24 Gag protein" (PMID: 30951545). Functionally, PEG10 **preferentially binds and**

facilitates vesicular secretion of its own mRNA (PMID: 34413232) and assembles into **virus-like particles (VLPs) that can be purified by iodixanol ultracentrifugation** (PMID: 39775359). Its RNA binding is sequence-selective: PEG10 "has selectivity for binding **U/G-rich RNAs**" (PMID: 42239172). In trophoblast stem cells, PEG10 also binds cellular transcripts (including *Hbegf*) and stabilizes mRNAs required for differentiation.

Together these data define the **primary molecular function**: PEG10 is a self-assembling, RNA-packaging **structural (capsid) protein** with sequence-selective RNA-binding activity.

Finding 3 — PEG10's POL-like viral aspartic protease is required for placental vascular maintenance and self-processing

PEG10 is translated as **GAG and GAG-POL** forms via a −1 ribosomal frameshift, and the POL-like region contains a conserved **viral aspartic protease (DSG/DTG) motif**. Knock-in mice carrying a point mutation in this motif survive early placentation but show **perinatal lethality with fetal/placental growth retardation and severe fetal capillary (endothelial) defects**; PEG10 is expressed in the "three trophoblast cell layers that surround fetal capillary endothelial cells" (PMID: 34559199). The protease is also central to PEG10's own biology: PEG10 "undergoes a complex self-processing pathway that results in gene expression changes when the protein accumulates in cells" (PMID: 36951542).

This finding dissociates two developmental requirements — the **Gag/capsid** function (needed for early placentation) and the **protease** function (needed later for fetal vasculature) — demonstrating that both retroviral-derived modules are physiologically active.

Finding 4 — PEG10 is an oncoprotein, notably in HCC, acting by stabilizing target mRNAs

PEG10 is overexpressed in HCC and associated with poor prognosis; liver-specific *Peg10* deletion impedes hepatocarcinogenesis in mice. Mechanistically, "as an RNA-binding protein (RBP), PEG10 inhibited the **Staufen1-mediated mRNA decay (SMD) pathway, stabilizing GLUT1 and HK2 mRNA** to enhance glucose uptake and aerobic glycolysis" (PMID: 42826094). This directly links PEG10's RNA-binding capsid function to a metabolic (Warburg-effect) oncogenic program. PEG10 protein abundance is tightly controlled: the E3 ligases SIAH1/SIAH2 "bound to the PAIR domain of PEG10 and promoted its polyubiquitination via K48/63-linked chains" (PMID: 42168988), while USP9X deubiquitinates and stabilizes PEG10 (PMID: 30951545). Transcriptionally, "CTR9 promoted the oncogene paternally expressed gene 10 (PEG10) transcription via its promoter region" (PMID: 34876700).

Finding 5 — In neurons, PEG10 is restrained by UBQLN2 and drives TDP-43-like splicing dysregulation

The proteasome shuttle factor **UBQLN2** restrains PEG10 to maintain neuronal health; ALS-linked UBQLN2 mutations permit PEG10 accumulation (PMID: 36951542). Excess PEG10 is functionally pathogenic: "PEG10 overexpression mimics the **loss of TDP-43** in broad changes to gene expression, including **dysregulation of mRNA splicing pathways**," with PEG10 binding U/G-rich RNAs (PMID: 42239172). PEG10 "has been implicated in neurological disease" (PMID: 39775359), and the related Gag-like gene **RTL8 antagonizes PEG10 VLP formation**, suggesting a built-in counter-regulatory mechanism.

Finding 6 — PEG10 (SIRH1) has a GAG-then-GAG-POL architecture generated by programmed −1 ribosomal frameshifting

PEG10 is the founding member (**SIRH1**) of the **sushi-ichi-related retrotransposon homolog (SIRH)** family of LTR-retrotransposon-domesticated genes that emerged with viviparous mammals. These genes "encode proteins that are homologous to a retrotransposon Gag capsid protein and sometimes also have a Pol-like region," and "PEG10 is conserved in both the marsupials and the eutherians" (PMID: 22866050). Like retroviruses, PEG10 produces a short ORF1 (GAG) and, via a conserved **GGGAAAC slippery site plus downstream RNA pseudoknot**, a **programmed −1 ribosomal frameshift (−1 PRF)** yielding the GAG-POL fusion containing the aspartic protease. This −1 PRF element is a defined, conserved translational-recoding mechanism "through RNA structural elements, most notably stimulatory pseudoknots" and can be selectively modulated by structure-guided antisense oligonucleotides (PMID: 42818233).

Finding 7 — PEG10 functions extracellularly via secreted VLPs that transfer mRNA; dysregulation disrupts TGF-β signaling in preeclampsia

PEG10 acts not only intracellularly but **extracellularly**. In the placenta, "PEG10 forms virus-like particles, which are **transferred from the trophoblast to the closely proximate endothelial cells**," and "excessive PEG10 disrupts maternal vascular development by **inhibiting TGF-β signaling**" in preeclampsia (PMID: 38715110). The programmable nature of the cargo — "the mRNA cargo of PEG10 can be reprogrammed by flanking genes of interest with [PEG10 UTRs]" (PMID: 34413232) — underpins the engineered **SEND** mRNA-delivery platform, and PEG10 eVLPs have been engineered to package neoantigen mRNA/peptides for cancer vaccines (PMID: 39269893).

This finding establishes PEG10 as a genuine **intercellular communication device** whose physiological output converges on the TGF-β pathway.

Finding 8 — PEG10 is an anti-apoptotic, proliferation-promoting protein that antagonizes SIAH1

The first functional characterization identified PEG10 as highly expressed in the majority of HCCs but absent in normal liver; "PEG10 protein associated with **SIAH1, a mediator of apoptosis**, and overexpression of PEG10 decreased the cell death mediated by SIAH1," while "exogenous expression of PEG10 conferred oncogenic activity and transfection of hepatoma cells with antisense S-oligonucleotides suppressing PEG10 resulted in their growth inhibition" (PMID: 12810624). Reciprocally, "SIAH1 also reduces the amount of PEG10 protein, which is known to be frequently overexpressed in HCC and to promote cell proliferation" (PMID: 17273732). PEG10-RF1 is thus an **inhibitor of apoptosis that mediates cell proliferation** (PMID: 20084274).

Finding 9 — Gene organization, two frameshift isoforms, cytoplasmic localization, and broad oncofetal overexpression

PEG10 maps to the **7q21 imprinted locus** (adjacent to *SGCE*) and encodes at least two proteins: **PEG10-RF1 (GAG-like)** and **PEG10-RF1/2 (GAG-POL)**, produced "by −1 frameshift translation," with "translation initiated at a **non-AUG start codon** upstream of the previously predicted AUG codon as well as at the AUG codon" (PMID: 20084274). PEG10 is a **cytoplasmic** protein that buds from membranes to form secreted VLPs (PMID: 30951545). Its oncofetal reactivation extends beyond HCC: in high-risk B-cell chronic lymphocytic leukemia, "expression levels in CD19+ B-CLL cells were **100-fold higher** than in B-cells from healthy donors," where it is prognostic and long-term knockdown induces apoptosis (PMID: 17621626).

Finding 10 — X-ray crystallography confirms PEG10 adopts a retroviral capsid (CA) fold

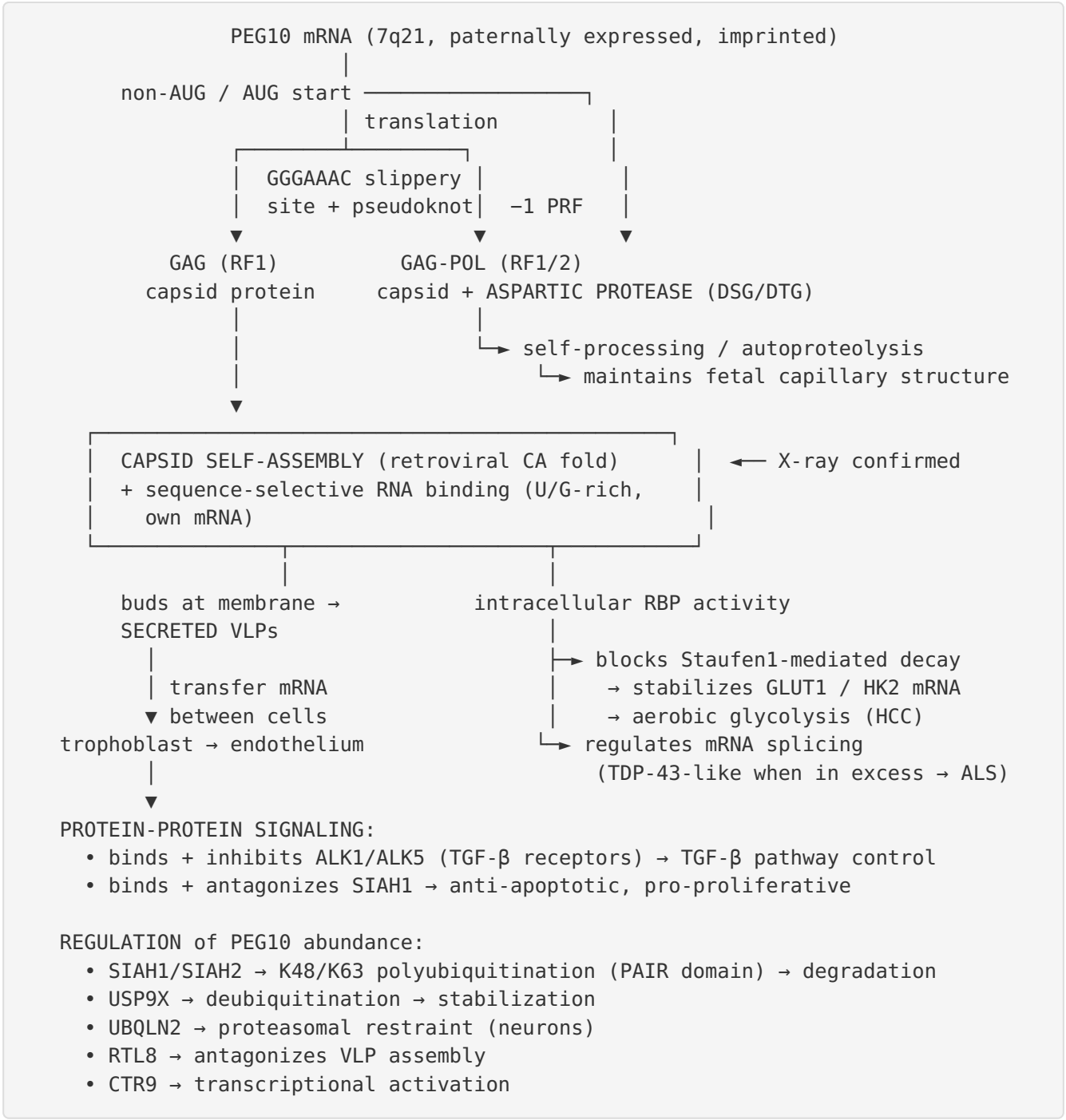
Direct structural evidence validates the Gag classification: "X-ray crystal structures of stably folded domains of MOAP1 ... and PEG10 ... are **highly similar to the C-terminal capsid (CA) domains of cognate Gag proteins**," and these structures "suggest that these proteins may have **preserved some of the key interactions that facilitated assembly of their ancestral Gags into capsids**" (PMID: 34357660). This provides structural (not merely sequence-based) support for the InterPro **PEG10-RTL1_C-capsid** domain (IPR063300) listed in the UniProt annotation.

Finding 11 — PEG10-RF1 directly binds and inhibits TGF-β superfamily type I/II receptors (ALK1, ALK5)

In a yeast two-hybrid screen for binding partners of **ALK1** (a TGF-β type I receptor; ACVRL1), **PEG10-RF1 was identified as an ALK1-binding protein**. "PEG10-RF1 also interacts with different members of TGF-β superfamily type I and II receptors. PEG10-RF1 binding to ALK1 is mediated by a **200-amino-acid domain with no recognized motif**. PEG10-RF1 **inhibits ALK1 as well as ALK5 signaling**" (PMID: 15611116). PEG10 was identified as "an ALK1-binding protein encoded by an ancient retroviral/retrotransposon element integrated as a single copy gene known as PEG10 on human chromosome 7q21," with broad expression "in ... human placenta, brain, kidney, endothelial cells, lymphoblasts." This provides the **molecular mechanism** connecting PEG10 to the TGF-β pathway modulation observed physiologically in preeclampsia (Finding 7).

Mechanistic Model / Interpretation

PEG10 is best understood as a **domesticated retrovirus-in-miniature** whose ancestral Gag/Pol biochemistry has been repurposed for mammalian physiology. The findings cohere into the following integrated model:



Three functional modules operate in parallel:

1. **Structural / RNA-packaging module (primary molecular function).** The Gag-derived capsid domain adopts a bona fide retroviral CA fold (Finding 10), self-assembles into VLPs (Findings 2, 9), and binds RNA with U/G-rich selectivity, preferentially its own mRNA (Finding 2). This makes PEG10 a **cytoplasmic/membrane-associated RNA-binding structural protein** that can bud into **secreted, mRNA-carrying particles** (Finding 7).

2. **Enzymatic module.** The −1 frameshift produces a GAG-POL fusion carrying a functional **aspartic protease** (Findings 3, 6) that mediates self-processing and is specifically required for fetal placental vasculature (Finding 3).
3. **Signaling/adaptor module.** PEG10-RF1 directly binds and inhibits **TGF-β receptors ALK1/ALK5** (Finding 11) and antagonizes the pro-apoptotic ligase **SIAH1** (Finding 8). These protein–protein interactions explain both its physiological role in vascular remodeling (TGF-β) and its anti-apoptotic/oncogenic activity.

Localization. PEG10 carries out its functions principally in the **cytoplasm** (RNA binding, budding at membranes, SIAH1/receptor interactions) and **extracellularly**, via secreted VLPs that deliver mRNA to neighboring cells (Findings 7, 9).

Physiology vs. pathology. The same machinery that builds the placenta (intercellular mRNA transfer + TGF-β tuning + anti-apoptosis) becomes pathogenic when dysregulated: **too much PEG10** disrupts maternal spiral-artery remodeling (preeclampsia), drives oncofetal cancers (HCC glycolysis via GLUT1/HK2 stabilization; B-CLL), and — when it escapes UBQLN2 control in neurons — produces TDP-43-like splicing collapse (ALS). This explains why PEG10 abundance is policed by a dedicated network (SIAH1/2, USP9X, UBQLN2, RTL8).

Summary table of PEG10 functional modules

Module	Domain / isoform	Molecular activity	Evidence type	Key PMIDs
Capsid / RNA packaging	GAG (RF1)	Self-assembly into VLPs; U/G-rich & self-mRNA binding; membrane budding	Experimental + X-ray structure	30951545, 34413232, 39775359, 42239172, 34357660
Protease	GAG-POL (RF1/2), DSG/DTG	Aspartic protease; self-processing; fetal capillary maintenance	Knock-in mouse; biochemistry	34559199, 36951542, 22866050
TGF-β signaling adaptor	RF1, ~200-aa domain	Binds/inhibits ALK1 & ALK5; disrupts vascular remodeling	Y2H + functional assays; in vivo	15611116, 38715110
Anti-apoptotic adaptor	RF1 (PAIR domain)	Binds/antagonizes SIAH1; promotes proliferation	Co-IP; overexpression/knockdown	12810624, 17273732, 20084274
Translational control	mRNA (slippery site + pseudoknot)	Programmed −1 frameshift; non-AUG + AUG initiation	Reporter/structure studies	22866050, 20084274, 42818233

Evidence Base

PMID	Title (abbrev.)	Contribution
16341224	*Deletion of Peg10 causes early embryonic lethality*	Genetic proof PEG10 is essential for placenta; Ty3/gypsy origin
34559199	*PEG10 viral aspartic protease domain essential for fetal capillary structure*	Protease motif required for placental vasculature; trophoblast localization
30951545	*Gag protein PEG10 binds RNA and regulates trophoblast stem cells*	Gag-domain budding like HIV p24; RNA binding; USP9X stabilization
34413232	*PEG10 packages its own mRNA; pseudotyped for delivery*	Selective self-mRNA packaging; programmable cargo (SEND)
39775359	*RTL8 antagonizes PEG10 VLPs*	VLP purification; neurological disease link; RTL8 counter-regulation
42239172	*PEG10 is a regulator of mRNA splicing in neurons*	U/G-rich RNA selectivity; TDP-43-like splicing dysregulation
22866050	*Genes domesticated from LTR retrotransposons (review)*	SIRH1/family architecture; GAG+POL; therian conservation
42818233	*Structure-guided ASOs modulate frameshifting*	−1 PRF/pseudoknot mechanism; therapeutic modulation
38715110	*Disruption of maternal vascular remodeling in preeclampsia (Nature)*	VLP transfer trophoblast→endothelium; TGF-β inhibition in vivo
15611116	*Gag/Gag-Pol-like proteins interact with ALK1*	Direct binding/inhibition of ALK1 & ALK5 (TGF-β receptors)
12810624	*PEG10 in HCC through interaction with SIAH1*	First functional study; anti-apoptotic, oncogenic; SIAH1 binding
17273732	*SIAH1 causes growth arrest/apoptosis in hepatoma*	PEG10 is a SIAH1 degradation target; pro-proliferative
20084274	*Genetic/molecular analyses of PEG10*	Two frameshift isoforms; non-AUG initiation; 7q21 locus
17621626	*PEG10 overexpression in high-risk B-CLL*	~100-fold reactivation beyond HCC; prognostic; apoptosis on knockdown
34357660		

PMID	Title (abbrev.)	Contribution
	Structural evidence MOAP1/PEG10 derived from Gag	X-ray CA fold; preserved capsid-assembly surfaces
42826094	*PEG10 inhibits mRNA decay to promote glucose metabolism in HCC*	Staufen1-mediated decay block; GLUT1/HK2 stabilization
42168988	*Polyubiquitination of PEG10 by SIAH1/2 in HCC*	PAIR-domain K48/K63 ubiquitination controls PEG10 abundance
34876700	*CTR9 promotes HCC via PEG10 transcription*	Transcriptional driver of PEG10 oncogenesis
36951542	*UBQLN2 restrains PEG10 in ALS*	Proteostatic control; self-processing; neurodegeneration link
39269893	*Engineering PEG10 eVLPs for cancer vaccination*	Applied proof of self-assembly/cargo-loading

Convergence of evidence types. The PEG10 model rests on an unusually strong, multi-modal evidence base: **genetics** (knockout and protease knock-in mice; Findings 1, 3), **structural biology** (X-ray capsid fold; Finding 10), **biochemistry** (RNA-binding selectivity, VLP purification, frameshifting assays; Findings 2, 6), **interaction studies** (Y2H and co-IP for ALK1/ALK5 and SIAH1; Findings 8, 11), and **in vivo physiology/pathology** (preeclampsia, HCC, ALS; Findings 4, 5, 7). These independent lines agree, giving high confidence in the core mechanistic model.

Limitations and Knowledge Gaps

- **Precise in vivo RNA cargo spectrum is incomplete.** While self-mRNA preference and U/G-rich selectivity are established (PMID: 34413232, PMID: 42239172), the full endogenous set of transcripts packaged and transferred by physiological PEG10 VLPs — and how this changes between placenta, neurons, and tumors — is not fully mapped.
- **Protease substrate specificity is largely undefined.** The aspartic protease is required for placental vasculature and self-processing (PMID: 34559199, PMID: 36951542), but the exact cleavage sites, kinetic parameters, and whether it has *trans* (non-self) cellular substrates remain open.

- **Mechanism of TGF-β receptor inhibition.** PEG10-RF1 binds/inhibits ALK1 and ALK5 via a ~200-aa domain with "no recognized motif" (PMID: 15611116); the structural basis and whether inhibition is competitive, allosteric, or trafficking-based is unknown.
- **Isoform-specific functions.** The division of labor between PEG10-RF1 (GAG) and RF1/2 (GAG-POL), and the regulation of the −1 frameshift ratio in different tissues, is not quantitatively resolved in vivo.
- **Human-specific vs. mouse data.** Several mechanistic experiments (protease knock-in, UBQLN2 restraint) rely on mouse models; human-specific validation is partial.
- **Causality in neurodegeneration.** The TDP-43-like effect is compelling (PMID: 42239172), but a direct causal demonstration that PEG10 accumulation initiates (rather than accompanies) human ALS pathology is still developing.
- **No new primary data generated here.** This report is a literature/annotation synthesis; findings reflect the cited studies' own designs, sample sizes, and caveats.

Proposed Follow-up Experiments / Actions

1. **Comprehensive VLP cargo mapping.** Perform CLIP-seq and VLP-RNA-seq on PEG10 particles purified from trophoblast, neuronal, and HCC cells to define the physiological, context-specific RNA cargo and test the U/G-rich/self-mRNA selectivity quantitatively.
2. **Protease substrate profiling.** Use N-terminomics/mass spectrometry with protease-dead (DSG/DTG mutant) controls to identify PEG10 autoproteolytic and potential *trans* substrates; determine cleavage consensus and kinetics.
3. **Structural characterization of the TGF-β interaction.** Solve a cryo-EM/X-ray structure of PEG10-RF1 bound to ALK1/ALK5 ectodomains or kinase domains to define the inhibitory interface and enable mechanism-based inhibitors.
4. **Quantify frameshift ratios in vivo.** Use targeted proteomics or frameshift reporters knocked into the endogenous locus to measure the GAG:GAG-POL ratio across placenta, brain, and tumor, and test whether structure-guided ASOs (PMID: 42818233) can therapeutically tune it.
5. **Targeting PEG10 in HCC.** Validate the Staufen1–GLUT1/HK2 axis (PMID: 42826094) as a metabolic vulnerability; test whether blocking PEG10 VLP assembly (e.g., via RTL8 mimetics; PMID: 39775359) suppresses glycolysis and tumor growth.

6. **Proteostatic rescue in neurons.** Test whether enhancing UBQLN2 function or SIAH-mediated degradation lowers neuronal PEG10 and reverses TDP-43-like splicing defects in patient-derived models (PMID: 36951542, PMID: 42239172).
7. **Preeclampsia biomarker/therapeutic.** Evaluate circulating PEG10 VLPs as a preeclampsia biomarker and test whether restoring TGF-β signaling counteracts PEG10-driven vascular defects (PMID: 38715110).

Conclusion

PEG10 (Q86TG7) is a **domesticated Ty3/gypsy retrotransposon-derived gene** whose primary molecular function is to act as a **retroviral Gag-like capsid protein that self-assembles into RNA-packaging virus-like particles** and, through programmed −1 frameshifting, additionally produces a **Gag-Pol form bearing a functional aspartic protease**. Operating in the **cytoplasm and extracellularly**, it transfers mRNA between cells, directly **inhibits TGF-β receptors ALK1/ALK5**, and **antagonizes the pro-apoptotic ligase SIAH1**. These activities make it essential and non-redundant for **placentation and placental-vascular maintenance**, while its dysregulation drives **oncofetal cancers** (notably HCC, via GLUT1/HK2 mRNA stabilization) and **neurodegeneration** (ALS/TDP-43-like splicing collapse when it escapes UBQLN2 control). The gene identity — symbol PEG10, human organism, and InterPro capsid/protease/reverse-transcriptase domains — is unambiguously confirmed by the literature synthesized here.
